

Student 3 - Implementation & Code Explanation

Time: 3 minutes

"Thank you. Now I will walk you through the actual code and how it works."

| Action: Open code editor or show code slides.

"We built the project in order. First Item.java - the data model. Then Validator. Then StorageManager. Then ItemService. Then Main. Finally Tests."

| Action: Show implementation order on screen.

"Let me show you how posting an item works. When you type 1 in the menu, Main.java calls postItem method."

| Action: Show Main.java postItem method on screen.

"The method asks for name, category, date, location, and contact. It validates that required

fields are not empty and date is in correct format."

| Action: Point to validation code lines.

"Then it calls ItemService.postItem. This loads all existing items, finds the highest ID, creates a new item with ID plus one, and saves it."

| Action: Show ItemService.postItem code.

"The save happens in StorageManager.append. It opens the file in append mode and writes one line. The line format is ID pipe type pipe name and so on."

| Action: Show the pipe-delimited format on screen.

"Now let me show you search. When you search by keyword, it loads all items and filters using Java streams. The filter checks if name, location, or category contains the keyword."

| Action: Show searchByKeyword code with stream pipeline.

"The claim process is interesting. It loads all items, finds the one with matching ID, changes

its status to RESOLVED, and rewrites the entire file."

| Action: Show claimItem method. Emphasize rewrite.

"We rewrite because every line contains the status field. When status changes, we must update the file completely."

| Action: Show file before and after claim.

"The data file uses pipe as delimiter. Each line has 8 fields separated by pipes. This makes it easy to split and parse."

| Action: Show example file line on screen.

"Error handling is built in. Missing files return empty lists. Bad input is caught and shown as friendly messages. The app never crashes."

| Action: Show error handling code examples.

"To compile, type javac dash d bin src slash star dot java. To run, type java dash cp bin Main."

| Action: Show compile and run commands.

"Now I will hand over for the live demonstration and testing."

| Action: Gesture towards next presenter. Step back.
